INCH-POUND

MIL-DTL-32326 (AR)
7 December 2009

DETAIL SPECIFICATION

FUZE, PYROTECHNIC SELF DESTRUCT - M1162

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and examination tests for the M1162 Pyrotechnic Self-Destruct Fuze (p-SDF) designed to minimize battlefield duds by adding self-destruct capability to the current Dual Purpose Improved Conventional Munitions (DPICM) M42/M46 grenades used in the Projectile, 155MM: Extended Range, Dual Purpose (DP), M864A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, NJ 07806-5000 or e-mailed to ardecstdzn@us.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1390

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-23132 - Tungsten Delay Composition

MIL-A-70625 - Automated Acceptance Inspection Equipment Design, Testing and Approval, of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test
MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

13001894 - Body Loading Assembly Type PH Inert Modified Studs
13001895 - Body Loading Assembly Type PH Modified Studs
13001896 - Body Loading Assembly Type CA Modified
13001897 - Body Loading Assembly Type CA Inert
13001899 - Body Fuze Assembly Type PH Inert
13001900 - Body Fuze Assembly Type PH
13001901 - Body Fuze Assembly Type CA Inert
13001902 - Body Fuze Assembly Type CA
13001903 - Grenade, GP: M46E3 Special, Inert
13001904 - Grenade, GP, M46E3
13001905 - Grenade, GP: M42E3 Special, Inert
13001906 - Grenade, GP M42E3
13001909 - Tape Loop for M42/M46 Grenade
13013438 - Interface Control Drawing, p-SDF

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13016740 - Fuze, General Purpose grenade: pSDF, M1162
13016741 - Slider Assembly
13016742 - Slider
13016743 - Ignitor
13016744 - Transfer Element
13016745 - Cap
13016746 - Housing
13016747 - Arming Screw Assembly
13016748 - Arming Screw
13016749 - Timing Plate
13016751 - Weight
13016753 - Safety Clip
13016756 - Delay Composition
13016758 - Inert Fuze, General Purpose Grenade: p-SDF, M1162
13016761 - Ignition Compound
13016762 - Ignitor Cup
13016764 - Trigger Assembly
13016768 - Intermediate Delay Composition
13016771 - Trigger
13016772 - Cover
13016773 - Slider Lock
13016779 - Trigger Weight
13016777 - Housing Assembly
9215255 - Safety Pin
9215360 - Arming Spring
8798331 - M55 Detonator

(Copies of these drawings may be requested on line at pica.drawing.request@conus.army.mil or from US Army ARDEC, RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample of fuze assemblies shall be subjected to the conformance inspection in accordance with 4.3.

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3.2 Components and assemblies. Materials, parts, and assemblies shall comply with drawings cited in 4.3.2.1 - 4.3.2.31.

3.3 M1162 pyrotechnic self destruct fuze (p-sdf) requirements.

3.3.1 Ballistic reliability. When fired from a 155MM howitzer, the combined primary mode and self destruct (M1162 fuze tactical mode) reliability of the p-SDF shall be not less than ninety nine (99) percent resulting in an Unexploded Ordnance (UXO) rate of not greater than one (1) percent when fired between the temperatures of -50°F and 145°F.

3.3.2 Ballistic safety reliability. There shall be no detonation of p-SDF in the gun tube or in flight prior to cargo expulsion and all M864A1 projectile hardware shall remain intact until initiation of the artillery projectile fuze.

3.3.3 Gun launch. The p-SDF shall not function in M864A1 projectile after being launched from a 155MM howitzer exerting a launch environment of 17,000 gravity and a spin of 10,173 Revolutions per Minute (RPM) to 16,374 RPM prior to initiation of the artillery projectile fuze.

3.3.4 Cargo ejection. The p-SDF shall function properly after cargo ejection at a maximum height of burst (HOB) of 1,800ft when expelled from an M864A1 projectile in-flight after initiation of the artillery projectile fuze.

3.3.5 Grenade loading. The p-SDF shall withstand the forces encountered during manual, semiautomatic, and fully automatic loading operations and shall be operable after being expelled from an M864A1 projectile in-flight after initiation of the artillery projectile fuze.

3.3.6 Primary arming torque. The arming screw unthreading torque shall be less than two (2) inch-ounce for individual fuze.

3.3.7 Primary arming. After unthreading the arming screw, the slider shall move to the armed position.

3.3.8 Tape loop pull-off. The tape loop shall withstand a pull-off force load of not less than 130 lbs.

3.3.9 Fuze pull-off. With the inert p-SDF mounted to a recapped inert armed body loading assembly, the fuze shall withstand a pull-off force of not less than 15,120 times the fuze weight.

3.3.10 Tape loop drop-off. The tape loop shall withstand a drop-off load of 1.00+ .25 lbs dropped from not less than three (3) inches.

3.3.11 Thermal shock. The p-SDF shall withstand sudden transitions in extreme low and high temperatures and shall comply with the Thermal Shock Test requirements of MIL-STD-331.

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3.3.12 Storage temperature. The p-SDF shall be safe for use and operable after being stored, handled, or transported in temperatures ranging between -65°F to 160°F.

3.3.13 Transportation vibration. The p-SDF shall be subjected to combined vibration-temperature environments anticipated during their logistical movement by land, air and sea conveyances and shall comply with the Transportation Vibration Test requirements of MIL-STD-331.

3.3.14 Temperature and humidity. The p-SDF shall withstand exposure to repeated cycles of extreme temperatures and humidity and shall comply with the Temperature and Humidity Test of MIL-STD-331.

3.3.15 Jolt. The p-SDF shall withstand a series of impacts applied in a controlled direction and amplitude, and shall comply with the Jolt Test requirements of MIL-STD-331.

3.3.16 Jumble. The p-SDF shall withstand random impacts imparted by free-fall inside a rotating, closed, lined box and shall comply with the Jumble Test requirements of MIL-STD-331.

3.3.17 Five foot drop. The p-SDF shall withstand several five foot drops onto a steel plate simulating accidental mishandling in manufacture and transportation, and shall comply with the Five Foot Drop Test requirements of MIL-STD-331.

3.3.18 Beam spring cocked pull force. The p-SDF beam spring shall withstand a pull force of not less than 1.5 lbs minimum when cocked.

3.3.19 Spring embrittlement. The spring shall not break or split when deformed to the minimum point of permanent distortion.

3.3.20 Safety pin embrittlement. The safety pin 9215255 shall not break or split when deformed to the minimum point of permanent distortion.

3.3.21 Slider assembly burn time. The slider assembly 13016741 shall pass a functional test after being conditioned at ambient, hot, and cold temperatures. The slider assembly shall burn between 12 to 16 seconds.

3.4 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.5 Workmanship. Molded component parts shall be free from splits, cold flow, shrinkage, flash, gouges, cold shuts, inclusion, porosity, or any similar characteristic. All parts shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light

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scratches not exposing base material are permitted. All required marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement and verification cross reference

Methods of Verification 1. Analysis 2. Demonstration 3. Examination 4. Test						Classes of Verification A. First Article B. Conformance Inspection		
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification Method
		1	2	3	4	A	B	
3.1.1	First article			x	x	x		4.2
3.1.2	Conformance inspection			x	x		x	4.3
3.2	Components and assemblies			x	x	x	x	4.3.2.1 – 4.3.2.28
3.3.1	Ballistic reliability			x	x	x	x	4.4.1
3.3.2	Ballistic safety reliability			x	x	x	x	4.4.2
3.3.3	Gun launch				x	x	x	4.4.3
3.3.4	Cargo ejection			x	x	x	x	4.4.4
3.3.5	Grenade loading			x	x	x	x	4.4.5
3.3.6	Primary arming torque				x	x	x	4.4.6
3.3.7	Primary arming				x	x	x	4.4.7
3.3.8	Tape loop pull-off				x	x	x	4.4.8
3.3.9	Fuze pull-off				x	x	x	4.4.10
3.3.10	Tape loop drop-off				x	x	x	4.4.9
3.3.11	Thermal shock				x	x		4.4.11
3.3.12	Storage temperature				x	x		4.4.12
3.3.13	Transportation vibration				x	x		4.4.13
3.3.14	Temperature and humidity				x	x		4.4.14
3.3.15	Jolt				x	x		4.4.15
3.3.16	Jumble				x	x		4.4.16
3.3.17	Five foot drop				x	x		4.4.17
3.3.18	Beam spring cocked pull force				x	x		4.4.18
3.3.19	Spring embrittlement				x	x		4.4.19
3.3.20	Safety pin embrittlement				x	x		4.4.20
3.3.21	Slider assembly burn				x	x		4.4.21-4.4.23
3.4	Ammunition lot numbering			x			x	4.4.30
3.5	Workmanship			x		x	x	4.4.31

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II.

4.2.1 First article quantity. First article inspection shall be performed on the quantities listed in Table II. For molded and stamped components, five (5) each shall be submitted from every die/mold cavity or stamping tool to be used in production.

4.2.2 Inspections to be performed. First article inspection shall be performed in accordance with Table II.

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TABLE II. First article inspection

Examination or test	Conformance criteria	Requirement paragraph	Inspection method reference
P-SDF, prior to attaching body loading assembly at grenade assembly point	10	3.2	4.3.2.1
Slider assembly	10	3.2	4.3.2.2
Slider	10	3.2	4.3.2.3
Ignitor	10	3.2	4.3.2.4
Transfer element	10	3.2	4.3.2.5
Cap	10	3.2	4.3.2.6
Housing assembly	10	3.2	4.3.2.7
Housing	See 4.2.1	3.2	4.3.2.8
Arming screw assembly	10	3.2	4.3.2.9
Arming screw	10	3.2	4.3.2.10
Timing plate	10	3.2	4.3.2.11
Weight	See 4.2.1	3.2	4.3.2.12
Safety clip	See 4.2.1	3.2	4.3.2.13
Arming spring	10	3.2	4.3.2.14
Safety pin	See 4.2.1	3.2	4.3.2.15
Ignitor cup	10	3.2	4.3.2.16
Trigger assembly	10	3.2	4.3.2.17
Trigger	10	3.2	4.3.2.18
Cover	See 4.2.1	3.2	4.3.2.19
Trigger weight	See 4.2.1	3.2	4.3.2.20
Delay composition	10	3.2	4.3.2.21
Ignition compound	10	3.2	4.3.2.22
Intermediate delay composition	10	3.2	4.3.2.23
Tape loop	10	3.2	4.3.2.24
Slider lock	See 4.2.1	3.2	4.3.2.25
Body loading assembly, modified studs	10	3.2	4.3.2.26
Grenade, GP, M42/M46 after fuze staking	10	3.2	4.3.2.27
Grenade assembly, prior to loading into projectile	/3	3.2	4.3.2.28

NOTES:
/1 Fuze testing may be combined per Figure 1 and 2 (see 6.17).
/2 100% of ballistic sample.
/3 See Figure 1 (see 6.17).
/4 See Figure 2 (see 6.17).
/5 See Figure 3 (see 6.17).

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TABLE II. First article inspection - Continued

EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Ballistic reliability test	2376 (33 projectiles)	3.3.1	4.4.1, /5
Ballistic safety test	216 (3 projectiles)	3.3.2	4.4.2, /5
Gun launch	216	3.3.3	4.4.3, /5
Cargo ejection	216	3.3.4	4.4.4, /5
Grenade loading	216	3.3.5	4.4.5, /5
Primary arming torque	10	3.3.6	4.4.6
Primary arming	10	3.3.7	4.4.7
Tape loop pull-off	10	3.3.8	4.4.8
Tape loop drop-off	10	3.3.10	4.4.9
Fuze pull-off	10	3.3.9	4.4.10
Thermal shock	40, /1	3.3.11	4.4.11, /4
Storage temperature	2376	3.3.12	4.4.12, /5
Transportation vibration	40, /1	3.3.13	4.4.13, /4
Temp and humidity	30, /1	3.3.14	4.4.14, /4
Jolt	10, /1	3.3.15	4.4.15, /3
Jumble	10, /1	3.3.16	4.4.16, /3
Five foot drop	50, /1	3.3.17	4.4.17, /3

NOTES:
/1 Fuze testing may be combined per Figure 1 and 2 (see 6.17).
/2 100% of ballistic sample.
/3 See Figure 1 (see 6.17).
/4 See Figure 2 (see 6.17).
/5 See Figure 3 (see 6.17).

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4.2.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The first article ballistic test shall be rejected if the sample experiences any of the following ballistic rejection criteria:

a. Any premature burst occurs.
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.
d. Unexploded ordnance on ground exceeds 1%.

4.3. Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with the lot formation requirement of MIL-STD-1916. In addition, inspection lots of finished projectiles shall contain:

a. Fuze, M1162 p-SDF, from one lot interfix number from one manufacturer.
b. Slider lock from one lot interfix number from one manufacturer.
c. Tape loop from one lot interfix number from one manufacturer.
d. Washer from one lot interfix number from one manufacturer.

4.3.2. Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.2.1 to 4.3.2.31, the definitions of critical, major and minor are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (see paragraph “Acceptance by contractor proposed provisions” of MIL-STD-1916).
d. Automated Acceptance Inspection Equipment (AAIE) shall be designed, qualified and calibrated in accordance with MIL-A-70625. AAIE shall automatically determine conformance of a characteristics without operator analysis or intervention.

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4.3.2.1	P-SDF prior to attaching body loading assembly at grenade assembly point.			Drawing number 13001898
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical				
1	Distance from bottom of cover to tip of arming screw, max.	100%, Level VII	3.2	AAIE
2	Safety clip missing	100%, Level VII	3.2	AAIE
3	Safety pin missing	100%, Level VII	3.2	AAIE
4	Trigger missing or out of position	100%, Level VII	3.2	AAIE
Major				
101	Cover fully seated on flange of fuze housing (4 places)	100%	3.2	Visual
102	Trigger weight assembly missing or improper	Level II	3.2	Gage
103	Safety pin height	Level IV	3.2	Gage
104	Fuze mounting hole diameters	Level II	3.2	Gage
105	Fuze mounting hole locations	Level II	3.2	Gage
106	Slider clearance to housing	Level IV	3.2	Gage
107	Spring coils extended beyond tab	100%	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.2	Slider assembly.			Drawing number 14016741
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016740 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	M55 detonator missing or improper	100%	3.2	X-ray
102	Improper detonator push-off	Level IV	3.2	Gage
103	Ignitor missing or improper	Level IV	3.2	X-ray
104	Epoxy in crossover hardness	Level IV	3.2	Gage
105	Safety pin pull out force	Level IV	3.2	Gage
106	Safety pin height	Level IV	3.2	Gage
107	Slider width	Level IV	3.2	Gage
108	Slider thickness	Level IV	3.2	Gage
109	Height of ignitor compound	100%	3.2	X-ray
110	Height of delay composition	100%	3.2	X-ray
111	Transfer element missing	100%	3.2	X-ray
112	Burn time	Level IV	3.2.21	4.4.21-4.4.23
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.3	Slider.			Drawing number 13016742
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness	Level IV	3.2	Gage
102	Width	Level IV	3.2	Gage
103	Safety pin hole diameter	Level IV	3.2	Gage
104	True position of safety pin hole	Level IV	3.2	Gage
105	Detonator hole diameter	Level IV	3.2	Gage
106	Detonator hole true position	Level IV	3.2	Gage
107	Arming screw hole diameter	Level IV	3.2	Gage
108	Arming screw hole true position	Level IV	3.2	Gage
109	Slider stop position and dimensions	Level IV	3.2	Gage
110	Diameter of ignitor leg	Level IV	3.2	Gage
111	True position of ignitor leg	Level IV	3.2	Gage
112	Diameter of transfer element leg	Level IV	3.2	Gage
113	True position of transfer element leg	Level IV	3.2	Gage
		Level IV	3.2	Gage
114	Diameter of ignitor hole	Level IV	3.2	Gage
115	True position of ignitor hole	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.4	Ignitor.			Drawing number 13016743
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Physical dimensions	Level IV	3.2	4.4.24
102	Function testing sensitivity	Level IV	3.2	4.4.25
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.5	Transfer element.			Drawing number 13016744
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical None defined				
Major 101	Physical dimensions	Level IV	3.2	4.4.26
Minor 201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.6	Cap.			Drawing number 13016745
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length, /1	Level IV	3.2	Gage
102	Width, /1	Level IV	3.2	Gage
103	Thickness, /1	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

NOTES:
/1 Measured as a flat blank.

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4.3.2.7	Housing assembly.			Drawing number 13016777
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016740 INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Correct assembly per drawing	Level IV	3.2	Visual
Minor 201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.8	Housing.			Drawing number 13016746
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016777 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fuze Mounting hole diameter (two places)	Level IV	3.2	Gage
102	Fuze Mounting hole true position (two places)	Level IV	3.2	Gage
103	Center hole diameter	Level IV	3.2	Gage
104	Center hole true position	Level IV	3.2	Gage
105	Housing lubricated	Level IV	3.2	Visual
106	Width of slider cavity	Level IV	3.2	Gage
107	Height of slider cavity tabs	Level IV	3.2	Gage
108	Presence of burrs	Level IV	3.2	Visual
109	Width between trigger hinge slots	Level IV	3.2	Gage
110	Width of arming screw assembly cavity	Level IV	3.2	Gage
111	Hardness	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.9	Arming screw assembly.			Drawing number 13016747
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016777 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Arming screw turns freely	Level IV	3.2	Visual & Manual
102	Timing plate turns freely on arming screw	Level IV	3.2	Visual & Manual
103	Retaining clip present	Level IV	3.2	Visual
104	Proper assembly	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.10	Arming screw.			Drawing number 13016748
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY 13016747 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Stab pin Tip flat diameter	Level IV	3.2	Gage
102	Thread improper	Level IV	3.2	Gage
103	True position of threads	Level IV	3.2	Gage
104	Arming screw flange diameter	Level IV	3.2	Gage
105	Arming screw flange length	Level IV	3.2	Gage
106	Body diameter	Level IV	3.2	Gage
107	Height from flange to top of thread	Level IV	3.2	Gage
108	Tip length	Level IV	3.2	Gage
Minor				
201	Tip length	Level II	3.5	Gage
202	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.11	Timing plate.			Drawing number 13016749
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016747 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width	Level IV	3.2	Gage
102	Length	Level IV	3.2	Gage
103	Inner diameter	Level IV	3.2	Gage
104	True position of inner diameter	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.12	Weight.			Drawing number 13016743
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016747 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Threads improper	Level IV	3.2	Gage
102	Width	Level IV	3.2	Gage
103	Length	Level IV	3.2	Gage
104	Top inner diameter	Level IV	3.2	Gage
105	Top outer diameter	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.13	Safety clip.			Drawing number 13016753
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016740 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Bend profile	Level IV	3.2	Gage
102	Thickness of material	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.14	Arming spring.			Drawing number 9215360
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016777 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Spring diameter	Level IV	3.2	Gage
102	Solid length (drawing note 3)	Level IV	3.2	Gage
103	Full length at load per (drawing note 13)	Level IV	3.2	Gage
104	Spring embrittlement	Level IV	3.2	4.4.19
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.15	Safety pin.			Drawing number 9215255
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016740 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length	Level IV	3.2	Gage
102	Diameter	Level IV	3.2	Gage
103	Pin embrittlement	Level IV	3.2	4.4.20
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.16	Ignitor cup.			Drawing number 13016762
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016743 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wall thickness	Level IV	3.2	Gage
102	Thickness of coined area at bottom of cup	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.17	Trigger assembly.			Drawing number 13016764
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016777 INSPECTION METHOD REFERENCE
Critical Major 101	None defined Weight push off force	 Level IV	 3.2	 Gage
Minor 201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.18	Trigger.			Drawing number 13016771
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016764 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Angle of point (70°)	Level IV	3.2	Gage
102	Profile of shark fin (R.400)	Level IV	3.2	Gage
103	Width of pivot arms	Level IV	3.2	Gage
104	Hardness	Level IV	3.2	Gage
105	Lubrication missing	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.19	Cover.			Drawing number 13016772
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016740 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of mounting holes	Level IV	3.2	Gage
102	True position of mounting holes	Level IV	3.2	Gage
103	Beam spring tip profile	Level IV	3.2	Gage
104	Beam spring pull force while cocked	Level IV	3.3.18	4.4.18
105	Hardness	Level IV	3.2	Gage
106	Thickness	Level IV	3.2	Visual
107	Lubrication missing	Level IV	3.2	Gage
108	Height of arming spring restraining feature (.090)	Level IV	3.2	Gage
109	Height of slider locking feature	Level IV	3.2	Gage
110	Spring retaining tab on arming spring feature missing (.056)	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.20	Trigger weight.			Drawing number 13016774
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016764 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total length	Level IV	3.2	Gage
102	Width	Level IV	3.2	Gage
103	Length of staking material	Level IV	3.2	Gage
104	Width of staking material	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.21	Delay compound.			Drawing number 13016756
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Delay compound composition	Level IV	3.2	4.4.27
102	Burn rate	Level IV	3.2	4.4.27.1
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.22	Ignition compound.			Drawing number 13016761
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ignition compound composition	Level IV	3.2	4.4.28
102	Burn time	Level IV	3.2	4.4.28.1
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.23	Intermediate delay compound.			Drawing number 13016768
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 13016741 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Intermediate delay compound composition	Level IV	3.2	4.4.29
102	Burn time	Level IV		4.4.29.1
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.24	Tape loop.			Drawing number 13001909
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Tape overlap welded improperly	Level IV	3.2	Visual
102	Width	Level IV	3.2	Gage
103	Maximum overlap distance between ends	Level IV	3.2	Visual
104	Pull test on tape 1oop to check weld strength	Level IV	3.2	Gage
105	Thickness	Level IV	3.2	Gage
106	Length	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual
NOTES:				

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4.3.2.25	Slider lock.			Drawing number 13016773
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness	Level IV	3.2	Gage
102	Outside profile prior to forming	Level IV	3.2	Gage
103	Angle of slide lock prior to forming	Level IV	3.2	Gage
104	Inside radius	Level IV	3.2	Gage
105	Distance to bottom of slide lock	Level IV	3.2	Gage
106	Radius of slide lock tab (2 places)	Level IV	3.2	Gage
107	Radius of slotted tab (2 places)	Level IV	3.2	Gage
108	Total width	Level IV	3.2	Gage
109	Distance across tabs at bottom	Level IV	3.2	Gage
110	Distance to outside of slide lock	Level IV	3.2	Gage
111	Distance to end of slide lock	Level IV	3.2	Gage
112	Hardness at location noted on drawing	Level IV	3.2	Gage
113	Thickness of protective finish	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

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4.3.2.26	Body loading assembly type PH and CA modified studs (after M223 fuze removal).			Drawing number /1
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY /2 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Lead cup assembly damaged, split, cracked, missing or improperly assembled	100%	3.2	Gage
102	Lead above flush or below max depth	Level IV	3.2	Gage
103	Stud loose or free to rotate, after fuze removal	Level IV	3.2	Gage
104	Depth of drill	Level IV	3.2	Gage
105	Any body cracked, dented or distorted after fuze removal	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

NOTES:
/1 13001894, 13001895, 13001896, 13001897
/2 13001899, 13001900, 13001901, 13001902

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4.3.2.27	Grenade, general purpose, M46/ M42 after staking of fuze with tape loop and slider lock.			Drawing number /1
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY /2 INSPECTION METHOD REFERENCE
Critical				
1	Slide protruding excessively from fuze indicating armed or partially armed condition	100%, Level VII	3.2	AAIE
2	Slider lock, safety pin, safety clip missing	100%, Level VII	3.2	AAIE
3	Distance from bottom of cover to tip of arming screw, max.	100%, Level VII	3.2	AAIE
4	Trigger missing or out of position	100%, Level VII	3.2	AAIE
Major				
101	Arming screw not free to rotate prior to furling tape loop.	Level IV	3.2	Visual/Manual
102	Evidence of tape loop loose or incorrectly contained by slide lock.	Level IV	3.2	Visual
103	Arming screw clinch missing, incomplete or not properly centered	Level IV	3.2	Visual
104	Washer missing	100%	3.2	Visual
105	Fuze housing cracked or deformed	Level IV	3.2	Visual
106	Stud clinch missing.	Level IV	3.2	Visual
107	Primary arming torque	Level IV	3.3.6	4.4.6
108	Primary arming	Level IV	3.3.7	4.4.7
109	Tape loop pull	Level IV	3.3.8	4.4.8
110	Tape loop drop off	Level IV	3.3.10	4.4.9
111	Tape loop missing or improperly contained in slider lock	Level IV	3.2	Visual

NOTES:
/1 13001903, 13001904, 13001905, 13001906
/2 13001907, 13001908

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4.3.2.27	Grenade, general purpose, M46/ M42 after staking of fuze with tape loop and slider lock.			Drawing number /1
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY /2 INSPECTION METHOD REFERENCE
Major				
112	Fuze pull off	Level IV	3.3.8	4.4.10
113	Ballistic test	1296, /4	3.3.1	4.5.3, /3
114	Ballistic safety test	216, /5	3.3.2	4.5.3.1, /3
115	Ammunition lot number	Level IV	3.4	Visual
116	Depth of recapped stud	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.5	Visual

NOTES:
/1 13001903, 13001904, 13001905, 13001906
/2 13001907, 13001908
/3. See figure 4. See 6.17
/4 Unexploded ordnance on test field exceeding one percent (1%) shall be cause for lot rejection.
/5 One or more defects shall be cause for lot rejection.

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4.3.2.28	Grenade assembly, prior to loading into projectile.			Drawing number /1
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY /2 INSPECTION METHOD REFERENCE
Critical				
1	Slide protruding excessively from fuze indicating armed or partially armed condition	100%, Level VII	3.2	AAIE
2	Slide Lock missing or disturbed.	100%, Level VII	3.2	AAIE
3	Safety pin missing	100%, Level VII	3.2	AAIE
4	Trigger missing or out of position	100%, Level VII	3.2	AAIE
Major				
101	Tape loop incorrectly contained by slide lock.	Level IV	3.2	Visual
Minor	None defined			
NOTES: /1 13001903, 13001904,13001905, 13001906 /2 13001907, 13001908				

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4.4 Methods of inspection.

4.4.1 Ballistic reliability. Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred. The round shall be fitted with a M762A1 fuze. The M762A1 fuze setting and quadrant elevation (QE) of the weapon shall be adjusted to attempt to achieve ejection over a prepared impact area at the target height specified below. During any test phase the fuze setting and QE shall remain constant. The maximum ejection height shall be 1,800 feet, for any zone. A photograph shall be taken of the fuze setting for each round immediately prior to use and identified with the round number. Projectiles shall be fired from an M198 cannon. Gun tubes used for this test shall not have less than 25 percent remaining tube life (for wear) based on current gun tube condemnation criteria. The ballistic sample shall be rejected if UXO on proving ground is more than one (1) percent (see 6.12) and shall cause reject of the lot.

4.4.2 Ballistic safety reliability test. Projectiles shall be fired with the same method as 4.4.1 with the exception of the propellant charge. The charge at the proving ground shall be adjusted to reach approximately 60,300 psi with the projectile conditioned to the required temperatures. The ballistic safety test shall cause reject of the lot if projectile hardware does not remain intact until initiation of artillery projectile fuze (see 6.12).

4.4.3 Gun launch test. This test shall be done in conjunction with the ballistic safety test in 4.4.2. P-SDF functioning prior to initiation of artillery projectile fuze shall cause reject of the lot.

4.4.4 Cargo ejection test. This test shall be done in conjunction with the ballistic safety test in 4.4.2. Cargo failing to eject or does not function as intended after initiation of artillery projectile fuze shall cause reject of the lot.

4.4.5 Grenade loading. This test shall be done in conjunction with the ballistic safety test in 4.4.2. The field shall be inspected by EOD personnel for separation of the p-SDF from the Body Loading Assembly as well as p-SDF affixed to the Body Loading Assembly abnormally. A count as well as recorded observations shall be taken of anomalies.

4.4.6 Primary arming torque. The fuze shall be assembled with a primary tape loop and staked onto an inert grenade with recapped studs and mounted in a fixture that prevents the slide from arming without applying force to the slider. The safety clip and safety pin shall be pulled off (removed). The fuze on inert grenade with the fixture shall be suspended at the primary tape loop on a torque watch. The arming screw shall then be rotated out of the slide, and the unthreading torque shall be read from the torque watch. If reading from torque does not meet criteria, then the lot from production shall be rejected. The sample of fuzes used for this test shall be used for subsequent tests throughout the fuze pull-off test in 4.4.10.

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4.4.7 Primary arming. After completion of the arming torque test in 4.4.6, while maintaining the arming screw free of the slide, the fixture that prevents the slide from arming shall be removed permitting the slide to move to the armed position. Any fuze failing to move to the fully armed position shall be classified as failure and the lot from production shall be rejected. After observation for compliance with the slide arming requirements and while still maintaining the arming screw free of the slide, the slide shall be returned to the unarmed position. The safety pin and safety clip shall be properly reinserted. The sample being tested shall be split in half. One half will be used to test tape loop pull off (see 4.4.8), and the other shall be used to test tape loop drop off (4.4.9). The whole sample will then combine to complete fuze pull off test (see 4.4.10).

4.4.8 Tape loop pull-off. After completion of the primary arming test in 4.4.7, the fuze shall be suspended on the primary tape loop in a fixture. A pull off force in accordance with 3.3.8 shall be applied to attempt to separate the tape loop from the fuze. If tape loop pulls off at a force lower than what is required (see 3.3.8), then the lot from production shall be rejected.

4.4.9 Tape loop drop-off. After completion of the primary arming test of 4.4.7 the fuze shall be suspended on the primary tape loop in a fixture. A drop off load of 1.00 + 0.25 lb from a height of 3 inches minimum shall be applied to attempt to separate the tape loop from the fuze. If the sample being tested fails the test criteria, then the lot from production shall be rejected.

4.4.10 Fuze pull-off. After completion of the tape loop pull off test in 4.4.8 and tape loop drop off test in 4.4.9, the same fuzes shall be subjected to the following pull off test. For this test the tape loop shall be removed from the fuze without damaging the arming screw threads. The safety pin and safety clip, if present, shall be removed. The arming screw shall be unthreaded to the arm position. A fixture shall engage the arming screw and pull the grenade body on the other end in an attempt to separate the fuze from the grenade. The fuze to grenade assembly shall withstand a minimum pull force in accordance with 3.3.9. A force shall be applied to the fuze in the flight configuration. After application of the tensile force, the distance between the entire interface (i.e. designed contact area) of the fuze housing and the cover, shall be measured, with the cover in contact with the grenade body. The measurement shall not exceed 0.005 inch in any location. If the sample being tested fails the test criteria, then the lot from production shall be rejected.

4.4.11 Thermal shock test. The test shall be performed on unarmed fuze. The test shall be conducted in accordance with procedures outlined in Test C7 "Thermal shock" of MIL-STD-331. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section C7.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.12 Storage temperature. This test shall be performed in-conjunction with the ballistic reliability test in 4.4.2. The projectiles fired for the ballistic reliability test (see

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4.4.2) shall be conditioned at temperatures ranging between -65°F and 160°F. If projectile fails to function as intended, then test is rejected.

4.4.13 Transportation vibration. The test shall be performed on p-SDF in the unarmed position. The test shall be conducted in accordance with procedures outlined in Test B1.1 "Transportation vibration" of MIL-STD-331,. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section B.1.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.14 Temperature and humidity. The test shall be performed on unarmed fuze. The test shall be conducted in accordance with procedure outlined in Test C1 "Temperature and humidity" of MIL-STD-331. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section C.1.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.15 Jolt test. The test shall be performed on unarmed fuze. The test shall be conducted in accordance with procedure outlined in Test A1 "Jolt" of MIL-STD-331. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section A1.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.16 Jumble test. The test shall be performed on unarmed fuze. The test shall be conducted in accordance with procedure outlined in Test A2.1 "Jumble" of MIL-STD-331. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section A2.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.17 Five foot drop test. The test shall be performed on unarmed fuze. The test shall be conducted in accordance with procedure outlined in Test A4.1 "One and one half meter (five-foot) drop" of MIL-STD-331. Upon completion of test, p-SDF shall be examined as stated by the guidelines specified in section A4.3 "Criteria for Passing Test" of MIL-STD-331. If the sample being tested fails the test criteria of MIL-STD-331, then the lot from production shall be rejected.

4.4.18 Beam spring cocked pull force. The test shall be performed on unarmed fuze. The fuze shall be placed in a holding fixture and a force of 1.5 lbs shall be applied to cock the beam spring. If the beam spring fails meet the minimum applied force, then the sample being tested shall be rejected.

4.4.19 Spring embrittlement. The spring shall be stretched along its longitudinal axis until the spring no longer returns to its original length or profile. Any spring that breaks prior to this point shall be classed defective. Springs subjected to this test shall

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not be returned to the lot. If one or more springs fail the test, the batch of springs shall be rejected.

4.4.20 Safety pin embrittlement. The formed pull pin and pull wire shall be flexed to the point of minimum permanent deformation. The wire shall be flexed in a manner to straighten the component and the pin shall be flexed to bend the component until deforming pressure causes the pin to no longer return to its formed shape and the pin contains a permanent slight bend. Any component that breaks or splits prior to this condition shall be classed defective. Components subjected to this test shall not be returned to the lot. If one or more safety pins fail the test, the batch of safety pins shall be rejected.

4.4.21 Slider assembly burn time ambient. The slider assembly shall be conditioned at 70°F ± 5°F and shall remain at the required test temperature for a minimum of 2 hours before testing. The slider assembly shall be functioned while maintaining the temperature at 70°F ± 5°F. The required delay time for the slider assembly is 16 seconds minimum and the minimum value is obtained when 3σ is subtracted from the mean value when a sample of fuze is tested.

4.4.22 Slider assembly burn time hot. The slider assembly shall be conditioned at 145°F ± 5°F and shall remain at the required test temperature for a minimum of 2 hours before testing. The slider assembly shall be functioned while maintaining the temperature of 145°F ± 5°F. The required delay time for the slider assembly is 16 seconds minimum and the minimum value is obtained when 3σ is subtracted from the mean value when a sample of fuze is tested.

4.4.23 Slider assembly burn time cold. The slider assembly shall be conditioned at -50°F ± 5°F and shall remain at the required test temperature for a minimum of 2 hours before testing. The slider assembly shall be functioned while maintaining the temperature of -50°F ± 5°F. The required delay time for the slider assembly is 16 seconds minimum and the minimum value is obtained when 3σ is subtracted from the mean value when a sample of fuze is tested.

4.4.24 Ignitor dimensions. The ignitor shall be in accordance with drawing number 13016743 and be .105±.010 length and .119+.000/.119-.003 height. The ignitor energetic fill shall be flush to .010 below surface of ignitor.

4.4.25 Ignitor sensitivity. The ignitor shall initiate when a weight of .25±.01 ounce is dropped from a height of 3 inches onto firing pin. If ignitor fails to initiate, the batch shall be rejected.

4.4.26 Transfer element dimension. The transfer element shall be in accordance with drawing number 13016744 and be .159±.003 length and .123+.000/.123-.003 height. The ignitor energetic fill shall be flush to .020 below surface of ignitor.

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4.4.27 Delay compound composition. Verify that the energetic mix of the delay compound is in accordance with MIL-T-23132 as well as drawing 13016756.

4.4.27.1 Burn time. The delay compound shall be tested in an aluminum block with a .125 diameter hole, 1.00 inches deep and pressed at 10,000 PSI. The burn rate shall be in accordance with the following:

Burn rate (seconds/inch)	
Recommended	Target
24-30	28

4.4.28 Ignition compound composition. Verify that the energetic mix of the ignition compound is in accordance with MIL-T-23132 as well as drawing 13016761.

4.4.28.1 Burn time. The delay compound shall be tested in an aluminum block with a .125 diameter hole, 1.00 inches deep and pressed at 20,000 PSI. The burn rate shall be in accordance with the following:

Burn rate (seconds/inch)	
Recommended	Target
4-9	6

4.4.29 Intermediate delay compound. Verify that the energetic mix of the intermediate delay compound is in accordance with MIL-T-23132 as well as drawing 13016768.

4.4.29.1 Burn time. The delay compound shall be tested in an aluminum block with a .125 diameter hole, 1.00 inches deep and pressed at 20,000 PSI. The burn rate shall be in accordance with the following:

Burn rate (seconds/inch)	
Recommended	Target
12-20	18

4.4.30 Ammunition lot numbering. Verify that the ammunition lot numbering are prepared and applied to each lot in accordance with MIL-STD-1168.

4.4.31 Workmanship. Visually verify that all parts and assemblies meet the requirements of 3.5.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging

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activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this specification are intended to be used in the recapitalization of M864A1 Projectiles utilizing self destruct fuzes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Packaging requirements (see 5).
c. Requirements for submission of first article sample (see 3.1).
d. Government Furnished Equipment.
e. Requirements for submittal of Acceptance Inspection Equipment (see 6.3).
f. Requirements for ammunition lot numbering (in accordance with MIL-STD-1168).
g. Requirements for Certification of nondestructive test (NDT) personnel (see 6.4).
h. Requirements for submission of test data (see 6.5).

6.3 Submission of inspection equipment for designs approval. Submit equipment designs, as required, to Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000

6.4 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment should be qualified and certified in accordance with NAS-410. Qualification to other standards must be submitted to and approved by Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000.

6.5 Submission of test data. In addition to the normal distribution of records, when the M864 Projectiles are procured by the U.S. Army Tank Automotive Command, one (1) copy of all ballistic data and ammunition data cards should be forwarded to: Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000.

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6.6 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gagged to determine acceptability.

6.7 Hybrid lotting by zone weight. Projectiles falling out of the inspection lot criteria due to the zone weight restriction may be accumulated into a hybrid lot.

6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor is to describe them in a written proposal and should be prepared to assure that the effectiveness is equal to, or better than quality assurance provisions specified herein.

6.9 Submission of alternative quality conformance provisions. All contractor-proposed alternative quality conformance provisions should be submitted to the Government for evaluation/approval by the technical activity responsible for the preparation of this specification.

6.10 Critical characteristic controls. All production contracts will contain requirements for handling critical defects. As part of all proposals, suppliers will be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier's integrated management system.

6.11 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or item for delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.12 Ballistic evaluation. After firing, the projectiles should be evaluated for compliance with 3.3.1 and 3.3.2. A round-by-round accounting should be made for Unexploded Ordnance on the field.

6.12.1 Exclusion. If, for any reason, the proving ground considers that test conditions have affected all or part of the test results, the proving ground should declare the affected results invalid and perform new tests on replacement projectiles as required. The following are a list of known exclusions for guidance to the evaluator:

a. Low bursts. If any test projectile functions below the minimum burst height (900 feet) and the result of that firing would cause rejection based solely on grenade related defects, a replacement cartridge may be fired.

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b. Lost grenades. If any test projectile has more than eight (8) lost grenades and none of the found grenades show a condition for rejection, a replacement round may be fired.

c. Missing fuzes. Grenades which are recovered without fuzes should be counted as grenades that fail to function properly. If a single projectile has more than two grenades recovered without fuzes, the projectile is considered to have failed.

d. Air detonations. Grenade air detonations should be counted and recorded for informational purposes.

6.13 Ammunition lot numbers. Ammunition lot numbers requires ammunition data cards in accordance with MIL-STD-1168.

6.14 Definitions

6.14.1 Air Detonation: A grenade air detonation is defined as functioning of the grenade in the air due to any cause during the time interval after expulsion from the projectile and prior to impact on the prepared field.

6.14.2 Unexploded Ordinance (UXO): A UXO is defined as an item where any energetic material is still remaining in the M42 or M46 grenade or the fuze assembly.

6.14.3 Critical Characteristics:

Critical characteristics are physical or functional attributes of a system, item, assembly, subassembly, component or material that, in the event of a nonconformance, is likely to:

1. Cause death, disability, injury or occupational illness that may result in hospitalization for individuals handling, using, storing, maintaining or depending upon the product;

2. Cause severe environmental damage that violates law or regulation;

3. Prevent performance of a tactical function of a system, exposing the user or allies to unacceptable levels of operational safety risk; or

4. Prevent performance of a tactical function of a weapon system or major item, where repair or replacement costs exceed $200,000 in Fiscal Year 2000 dollars.

6.14.4 Automatic Acceptance Inspection Equipment (AAIE): Use of non-operator-dependent, repeatable, automatic-decision-making inspection equipment.

6.14.5 Dual-Purpose Improved Conventional Munitions (DPICM): Is an artillery or surface-to-surface missile warhead designed to burst into sub-munitions at an optimum altitude and distance from the desired target for dense area coverage. The sub-munitions are designed for both anti-armor and antipersonnel attack.

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6.15 Justification for critical characteristics.

6.15.1 Distance from bottom of cover to tip of arming screw, max. The location of the Arming screw tip is essential to prevent the slider from arming prior to intended function. Should the slider arm during loading or assembly an initiation of the detonator would function the grenade causing injury or death.

6.15.2 Safety pin missing. The safety pin is one of two main safety features whose presence is essential to maintain the slider in a safe position during manufacturing, shipping, and handling. Prior to loading the grenade into the projectile cavity, the probability of the slider on self destruct fuze moving to in-line firing position is increased if the safety pin is missing. Safety pin also holds the trigger assembly in place, preventing the beam spring tip from striking the ignitor. If the slider is in the in-line firing position, either the self destruct feature or the arming screw is able to initiate the detonator. This is critical because detonation in the armed position will result in unintentional function causing injury or death.

6.15.3 Safety clip missing. The safety clip prevents disengagement of the arming pin from the slider and is one of two main safety features whose presence is essential to maintain the slider in a safe position during manufacturing, shipping, and handling. Prior to loading the grenade into the projectile cavity, the probability of the slider on self destruct fuze moving to in-line firing position is increased if the safety clip is missing. If the slider is in the in-line firing position, either the self destruct feature or the arming screw is able to initiate the detonator. This is critical because detonation in the armed position will result in unintentional function causing injury or death.

6.15.4 Trigger missing or out of position. The beam spring is held in place by trigger assembly. A missing trigger could inadvertently cause the beam spring to strike the ignitor and set off the M55 detonator. A trigger out of position could inadvertently cause the beam spring to strike the ignitor and set off the pyrotechnic element and then the M55 detonator. The trigger missing, like the trigger out of place, is essential to prevent the igniter from functioning during manufacturing, shipping, and handling. The beam spring restrained by the trigger maintains safety until deployment. This is critical because unintentional functioning can cause injury or death.

6.15.5 Slide protruding excessively from fuze indicating armed or partially armed condition. It is essential for the slider to remain in the fully safe position to prevent grenade detonation other than as intended. This is critical because unintentional functioning can cause injury or death.

6.15.6 Slider lock missing. The slider and tape loop are held in place by the slider lock. A missing slider lock could inadvertently allow the slider to move to the armed position. This is critical because unintentional functioning can cause injury or death.

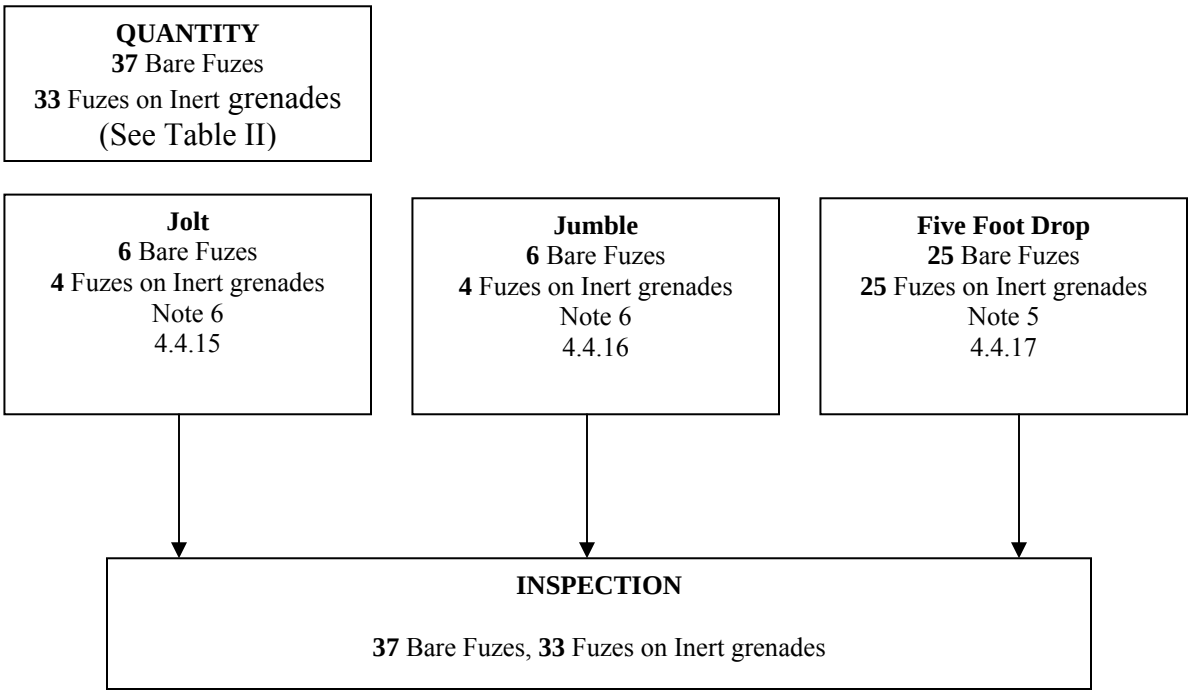
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6.16 Subject term (key) listing.

Gun Launch
Grenade
Unexploded Ordnance (UXO)

6.17 Figures. The below figures serve as an optional guide for the first article and lot acceptance tests:

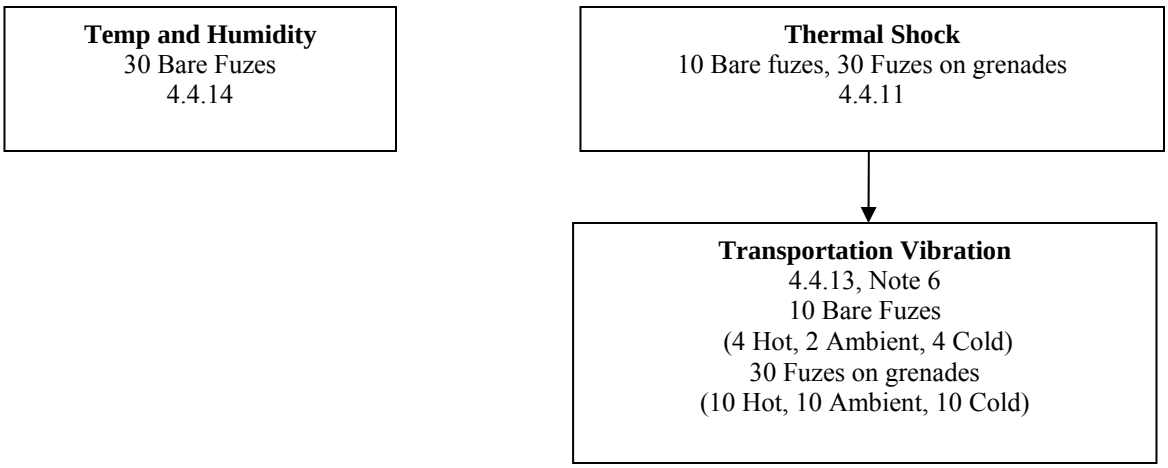
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NOTES:
1. All fuzes are tactical.
2. All fuzes on grenades shall be staked to inert grenades with recap studs.
3. All fuzes are tested with safety pin and safety clip unless specified differently.
4. Clearing of potential/growing failure mode is not allowed.
5. Fuzes on grenades shall have safety pin only.
6. Fuzes on grenades will be assembled in a housing representing projectile confinement without safety pin and safety clip (assembled with tape loop and tape loop retainer).
7. Inspect fuzes as necessary between test environments.
8. Bare fuzes in jolt and jumble tests represent plant-to plant transportation and shall be handled in packaged configuration.
9. Hot = 145°F, ambient = 70°F, cold = -50°F.

FIGURE 1. First article fuze verification test 1

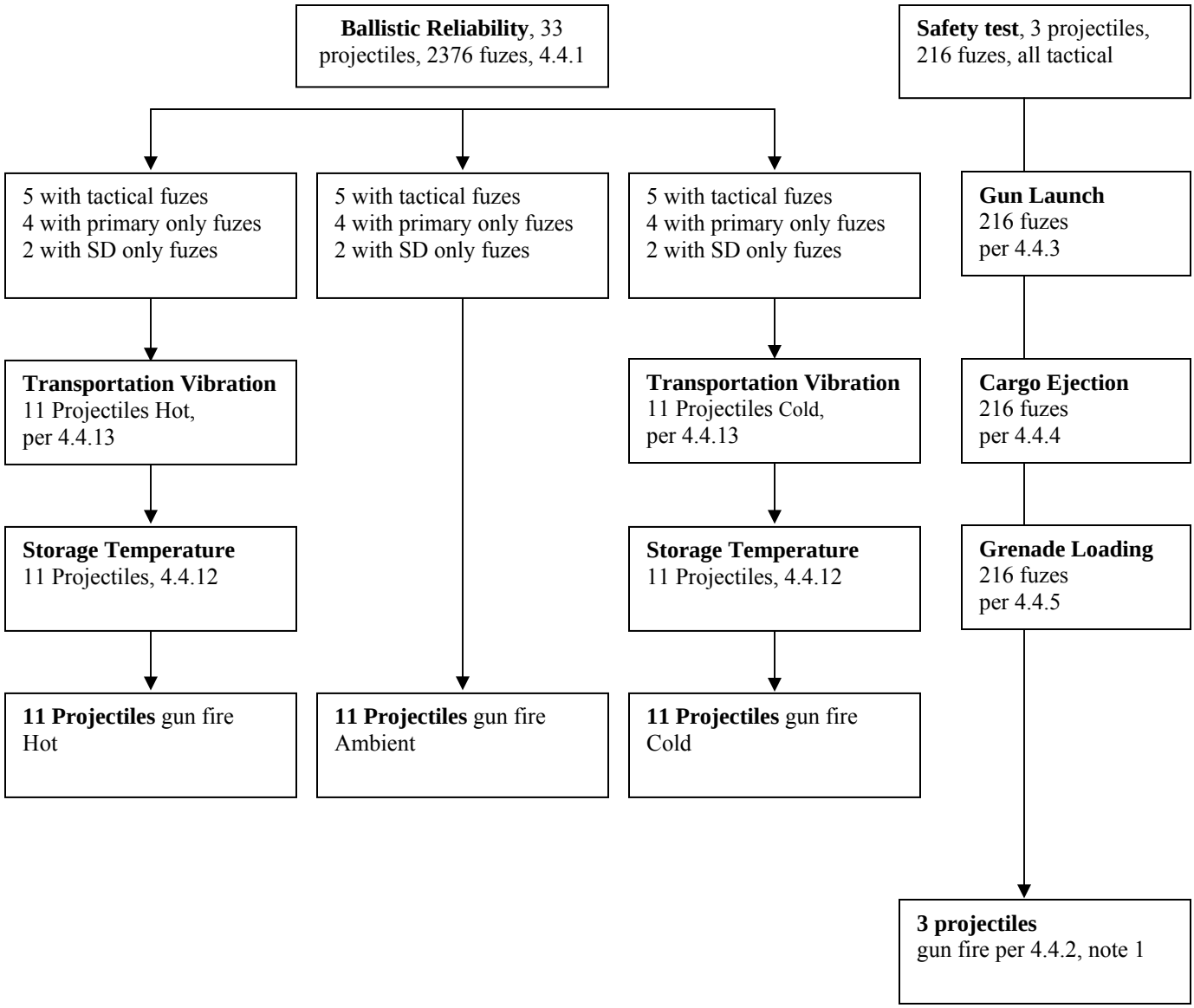
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NOTES:
1. All fuzes are tactical.
2. All fuzes on grenades shall be staked to inert grenades with recap studs.
3. All bare fuzes are tested with safety pins and safety clip unless specified differently.
4. Clearing of potential/growing failure mode is not allowed.
5. Fuzes on inert grenades will have safety pin only unless specified differently.
6. Fuzes on grenades will be assembled in a housing representing projectile confinement without safety pin and safety clip (assembled with tape loop and tape loop retainer).
7. If the gun launch shock and ground shock cannot be performed with fuzes staked to grenades then those fuzes may be run out of sequence, as bare fuzes prior to thermal shock and vibration with temperature. The 30 fuzes shall then be staked to grenades for the thermal shock and vibration.
8. Inspect between each environment.
9. Hot = 145°F ,ambient = 70°F ,cold = -50°F .

FIGURE 2. First article fuze verification test 2

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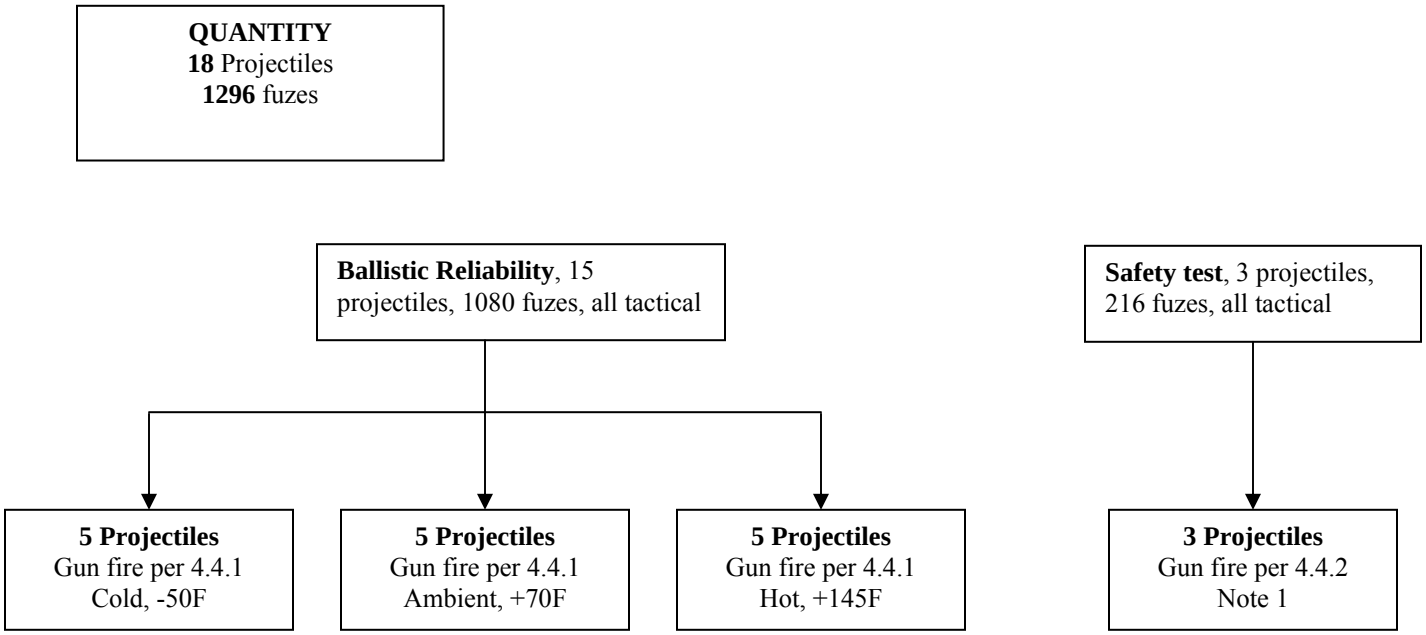


NOTES:

1. Condition round and propellant hot +145°F. Fire PIMP charge at approximately 60,000 psi.
2. All grenades are live.
3. Hot = 145°F , ambient = 70°F , cold = -50°F .

FIGURE 3. First article ballistic test

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NOTES:
1. Condition round and propellant hot +145F fire pimp charge at approximately 60,300 psi.
2. All grenades are live.

FIGURE 4. Lot acceptance ballistic test

Preparing activity:
Army-AR
(Project 1390-2009-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil

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